

Module 1: Introduction to Web Information Systems

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Learning Objectives

At the end of this module, students should be able to:

- Explain what an Information System is
- Describe how information systems evolved into web-based systems
- Identify the role of Web Information Systems in organizations and society

What is an Information System (IS)?

- A system that collects, processes, stores, and distributes information
- Supports decision-making, coordination, and control
- Combines **people, processes, data, and technology**

Components of an Information System

- Hardware
- Software
- Data
- Procedures
- People

Types of Information Systems

- Transaction Processing Systems (TPS)
- Management Information Systems (MIS)
- Decision Support Systems (DSS)
- Executive Information Systems (EIS)

Evolution to Web-Based Information Systems

Early Information Systems

- Standalone systems
- Mainframe computers
- Limited user access

Client/Server Systems

- Desktop applications
- Local Area Networks (LAN)
- Centralized databases

Internet and the World Wide Web

- Rise of the Internet
- HTTP and browsers
- Information accessibility

Timeline of IS Evolution

- 1960s–70s: Mainframe systems
- 1980s–90s: Client–server
- 2000s: Web-based systems
- 2010s–Present: Cloud and mobile systems

Why Organizations Shifted to Web IS

- Accessibility
- Cost efficiency
- Scalability
- Real-time data

Characteristics and Components of Web Information Systems

What is a Web Information System?

- An information system that uses web technologies
- Accessible via web browsers
- Supports business and organizational processes

Key Characteristics of Web IS

- Web-based access
- Distributed architecture
- Real-time processing
- Scalability
- Security considerations

Core Components of Web IS

- Client (browser)
- Web server
- Application server
- Database server

Supporting Technologies

- Internet and intranet
- Web servers (Apache, IIS)
- Databases (MySQL, PostgreSQL)
- APIs and web services

Sample Web IS Architecture

User → Browser → Web Server → Application
Logic → Database

Quick Discussion

- “What happens if the database server goes down?”

Thank You